APPLICATIONS OF DIFFERENTIATION

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Lecture Outline

1. L'hospital's Theorem
2. Critical Values
3. Minimum and Maximum Values
4. Mean value theorems
5. Shapes of graphs
6. Optimisation
7. Linear approximations
8. Small Variations



Outline of Presentation

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5. Limits of the form $\frac{0}{0}$ and $\frac{\infty}{\infty}$ are called **indeterminate** type.
6. To evaluate limits of indeterminate type we employ the L'Hospitals rule.



L'Hospital's rule: I

Definition

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \to a} f = 0 \quad \text{and} \quad \lim_{x \to a} g = 0$$

or

$$\lim_{x \to a} f = \pm\infty \quad \text{and} \quad \lim_{x \to a} g = \pm\infty$$

Then

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Note: Only apply L'Hospital's rule to indeterminate forms of limits.

Example

Evaluate $\lim_{x \to 1} \dfrac{\ln x}{x-1}$.



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Notice that $\lim_{x\to 1} \ln x = 0$ and $\lim_{x\to 1} x - 1 = 0$ hence the given limit is an indeterminate type. Thus

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Evaluate $\lim_{x\to\infty}\dfrac{e^x}{x}$.



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Example

Evaluate $\lim_{x\to \infty} \dfrac{e^x}{x}$.

$$\lim_{x\to \infty} \frac{e^x}{x} = \lim_{x\to \infty} \frac{e^x}{1} = \lim_{x\to \infty} e^x$$



Example

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Notice that $\lim_{x\to 1}\ln x = 0$ and $\lim_{x\to 1} x-1 = 0$ hence the given limit is an indeterminate type. Thus

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Example

Evaluate $\lim_{x\to\infty}\dfrac{e^x}{x}$.

$$\lim_{x\to\infty}\frac{e^x}{x} = \lim_{x\to\infty}\frac{e^x}{1} = \lim_{x\to\infty} e^x = \infty$$



L'Hospital's rule: II

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Example

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Solution

Notice that this indeterminate type. Thus

$$\lim_{x\to 0^+} x\ln x = \lim_{x\to 0^+} \frac{\ln x}{1/x} \tag{2}$$



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$$= 0 \tag{5}$$



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3. We simplify such cases into quotients and apply the L'Hospital's rule.



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3. We solve this by taking the natural logarithm or exponential and solving using the product case.



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Definition (Critical Point)

A critical Point of f is a number c such that either

1. $f'(c) = 0$ or
2. $f'(c)$ does not exist.



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Find the critical points of $1).\ f(x) = 10x^2 + 4x$ and $2).\ f(x) = 6x^5 + 33x^4 - 30x^3 + 100$



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$$= 6x^2(5x^2 + 22x - 15) \tag{13}$$

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$$= 6x^2(5x^2 + 22x - 15) \tag{13}$$

$$= 6x^2(5x - 3)(x + 5) \tag{14}$$

Equating (14) to zero, then the critical points are

$$x = 0 \qquad x = 3/5, \qquad x = -5 \tag{15}$$

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Minimum and Maximum Values

Our goal is to investigate the maximum and minimum values (extrema) of a function. Recall that

Definition

1. A function f has an absolute maximum (or global maximum) at **c** if $f(\mathbf{c}) \geq f(x)$ for all x in D_f . The number $f(\mathbf{c})$ is called the maximum value of on D_f.



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3. A function f has a local maximum (or relative maximum) at **c** if $f(\mathbf{c}) \geq f(x)$ for all x near **c**.



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To find the extrema of a function f on $[a, b]$

1. Find all the critical points of f
2. Determine the values of f at the critical points, and at a and b.
3. Read off the largest and minimum values as absolute maximum and minimum respectively.



Example

Find the absolute maximum(minimum) values of $f(x) = 3x^2 - 12x + 5$ on $[0, 3]$.



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The critical points is:

$$f'(x) = 6x - 12 \implies x = 2 \tag{16}$$

Evaluating

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1. $f(0) = 5$
2. $f(2) = -7$
3. $f(3) = -4$

Therefore, the absolute maximum is 5 and the minimum is -7

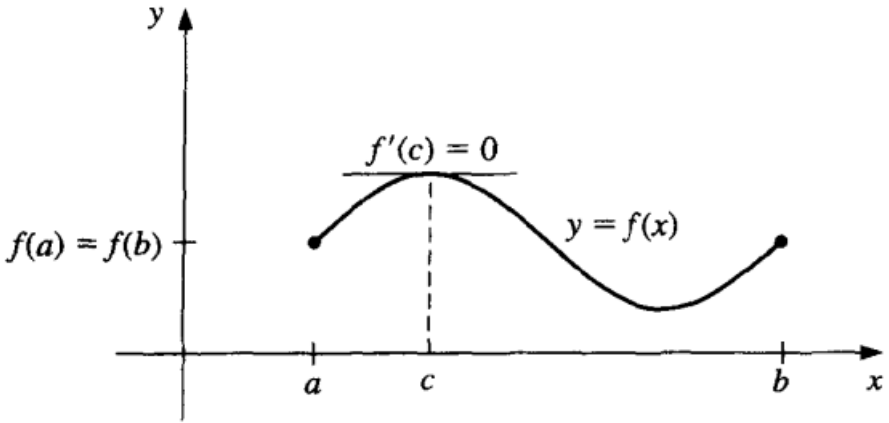


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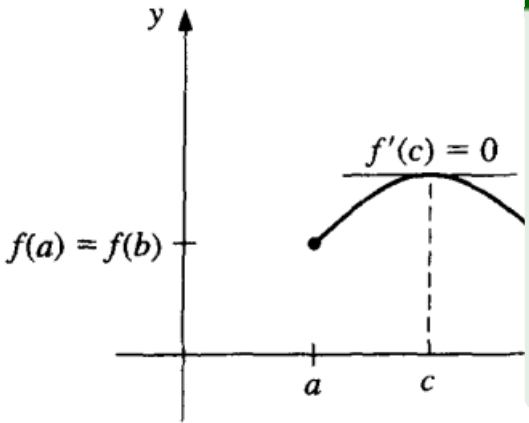
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Let f be a function such that

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b)
3. $f(a) = f(b)$
4. Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.

This indicates that f has an extrema value at a point of the interval $[a, b]$ if $f(a) = f(b)$.

Mean value theorems

Mean Value Theorem

Let f be a function such that

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b)
3. Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \qquad (17)$$

That is if $f'(c) = m$ then m is the mean value of f o $[a, b]$



Mean value theorems

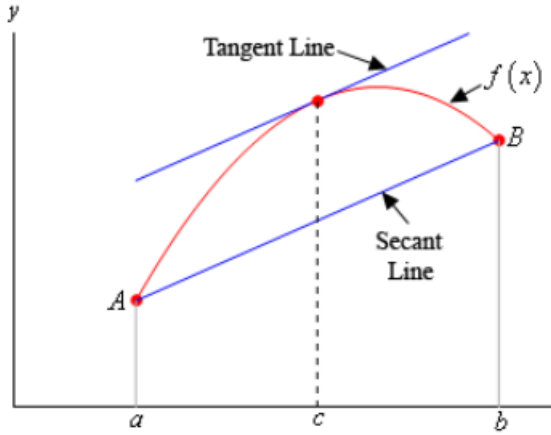
Mean Value Theorem

Let f be a function such that

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b)
3. Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \qquad (17)$$

That is if $f'(c) = m$ then m is the mean value of f o $[a, b]$



Example

Determine all the numbers c which satisfy the conclusions of the Mean Value Theorem for the following function.

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Solving

$$c = 0.7863 \qquad \text{and} \qquad c = -2.1196 \tag{21}$$

Notice that only one value is in the interval given, hence $c = 0.7863$



Remark

If $f'(x) = 0$ for all x in an interval (a, b), then $f(x)$ is constant on (a, b).



Outline of Presentation

1. L'hospital's Theorem
2. Critical Values
3. Minimum and Maximum Values
4. Mean value theorems
5. Shapes of graphs
6. Optimisation
7. Linear approximations
8. Small Variations



Shape of graphs: First Derivative

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Example

Determine all intervals where the following function is increasing or decreasing.

$$f(x) = -x^5 + \frac{5}{2}x^4 + \frac{40}{3}x^3 + 5$$

1. To determine if the function is increasing or decreasing we will need the derivative.

$$f'(x) = -5x^4 + 10x^3 + 40x^2 \tag{22}$$
$$= -5x^2(x^2 - 2x + 8) \tag{23}$$
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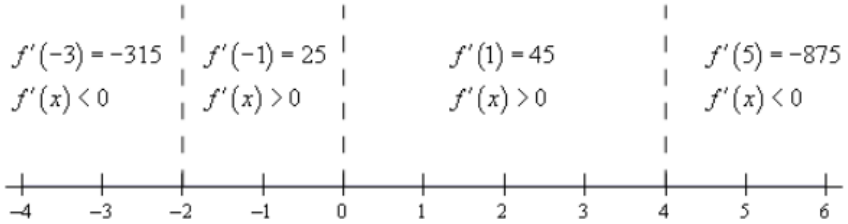
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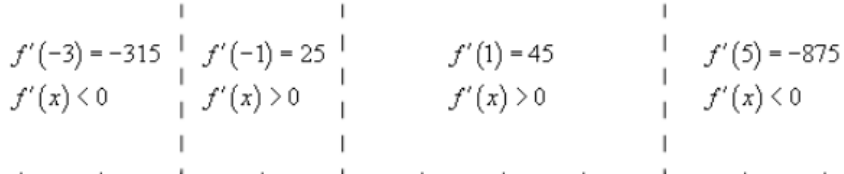


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4. So f is increasing on $(-2,\ 0)$ and $(0,\ 4)$ and decreasing on $(-\infty,\ -2)$ and $(4,\ \infty)$

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These ideas can be summarized up in the following test.

First Derivative Test

Suppose $x = c$ is a critical point of $f(x)$, then

1. If $f'(x) > 0$ to the left of $x = c$ and $f'(x) < 0$ to the right of $x = c$ then $x = c$ is a relative maximum



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Note: The first derivative test will only classify critical points as relative extrema and not as absolute extrema.

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Loosely, a function is concave up if it opens up and the function is concave down if it opens down.

Definition

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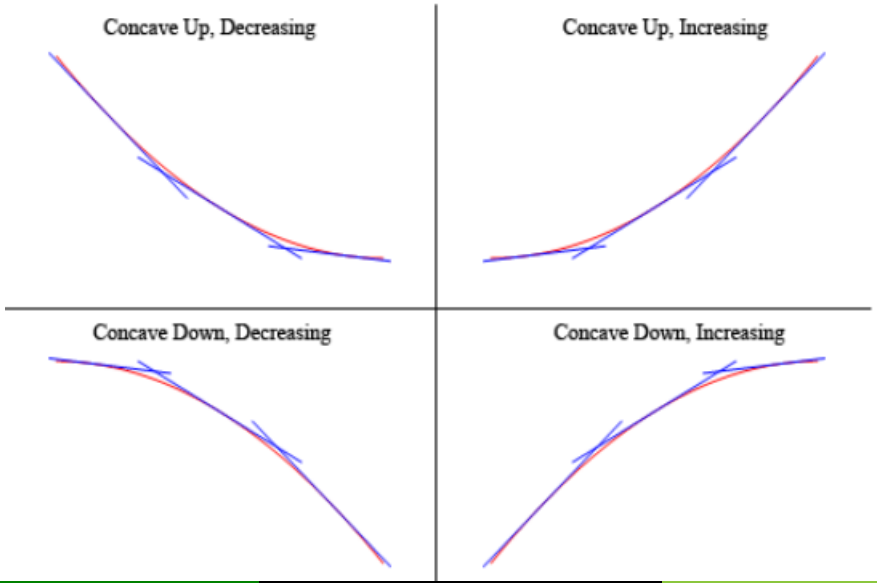
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Note

Concavity has nothing to do with increasing or decreasing. A function can be concave up and either increasing or decreasing. Similarly, a function can be concave down and either increasing or decreasing.


Concave Up, Decreasing
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Suppose f'' is continuous near a critical point c.

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3. If $f'(c) = 0$ and $f''(c) = 0$, then f can have a relative maximum, minimum or neither at c. This is called the point of inflection



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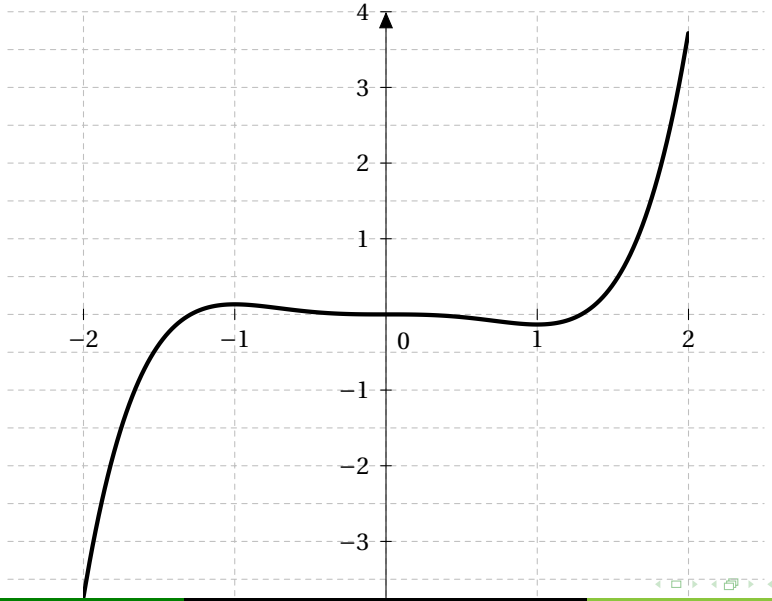


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4
3
2
1
−2
−1
0
1
2
−1
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3. Hence the dimension is 600×1200.



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3. Thus $V(1.2910) = 2.1517m^3$



Outline of Presentation

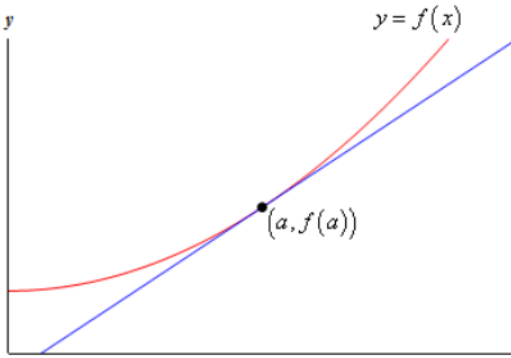
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Linear approximations

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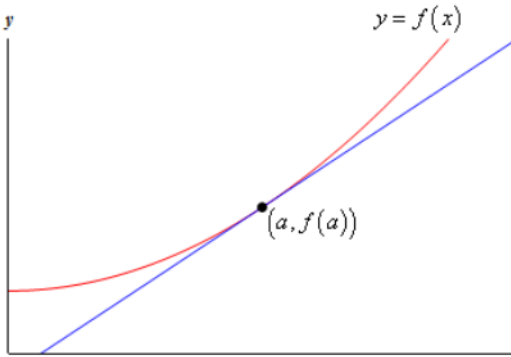
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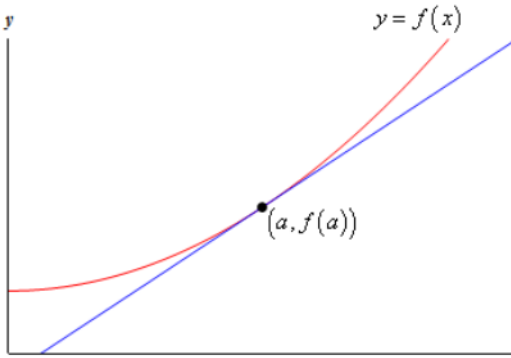


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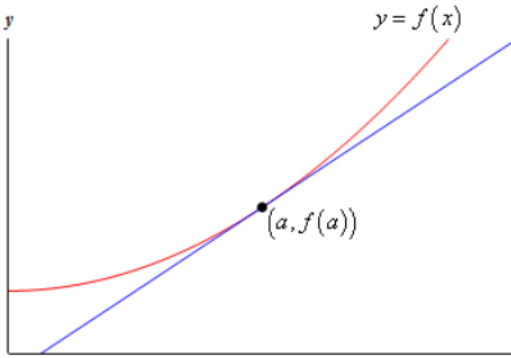
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Determine the linear approximation for $f(x) = \sqrt[3]{x}$ at $x = 8$. Use the linear approximation to approximate the value of $\sqrt[3]{8.05}$.



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There is no much difference between the exact solution 2.00415802 and its linear approximation 2.00416667.



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2. Typically, we identify the relationship between the quantities and use the chain rule to differentiate both sides with respect to time.



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$$\frac{dy}{dt} = -\frac{6}{8}(1) = -\frac{3}{4}ft/s$$



Exercise

1. A water tank has the shape of an inverted circular cone with base radius $2m$ and height $4m$. If water is being pumped into the tank at a rate of $2m^3/min$, find the rate at which the water level is rising when the water is $3m$ deep.
2. A cylindrical can is to be made to hold $1000cm^3$ of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.
3. Find the interval within which $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and decreasing.
4. Find the critical points, endpoints of D_f, the value of the function at each of these points and identify extreme values. $a)\ f(x) = x\sqrt{4-x^2}$, $b)\ f(x) = x^2\sqrt{3-x}$.
5. The height of a body moving vertically is given by $h(t) = -\frac{1}{2}gt + v_0t + h_0$ where $g > 0$. Find the body's maximum height in terms of h_0, v_0 and g.



Exercise

1. Does $f(x) = x^2 - 4x + 1$, $[0,4]$ satisfy the Rolle's theorem. If yes find c.
2. Does $f(x) = \dfrac{x^2 - x - 6}{x - 1}$, $[-2,4]$ satisfy the Rolle's theorem. If yes find c.
3. Does $f(x) = x^3 - x$, $[0,2]$ satisfy the Mean Value theorem. If yes find c.
4. Use $L(x)$ to linearize $g(x)$ in the following:
   a) $g(x) = (1-x)^6$
   b) $g(x) = \frac{2}{\sqrt{4+3x}}$
   c) $g(x) = \sqrt[3]{\left(1 - \frac{x}{2+x}\right)^2}$
5. The position at time t of a moving particle is given by $s(t)$. Find the time at which its instantaneous velocity $s'(t)$ is equal to its average velocity in the interval T.
   a) $s(t) = x^2 + 2x - 1, T = [0,1]$. *b)* $s(t) = t + \frac{1}{t}, T = [1/2, 2]$.
   c) $s(t) = \sqrt{t-1}, T = [1,3]$.



END OF LECTURE
THANK YOU

